

When Do Physics Priors Help?

A Controlled Benchmark of Physics-Informed Dynamics Models for Underactuated Swing-Up

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Abstract—Physics-informed machine learning promises data-efficient, physically consistent dynamics models, but it is often unclear *when* a physics prior actually pays off relative to a black-box baseline. We present a controlled, reproducible benchmark that compares five learned dynamics models—a vanilla multilayer perceptron (MLP), a Gaussian process (GP), a Lagrangian Neural Network (LNN), a Hamiltonian Neural Network (HNN), and a Lagrangian-informed GP (LGP)—on two classic underactuated systems, the Acrobot and the Pendubot. We evaluate along four axes: one-step prediction accuracy, long-horizon rollout stability (valid prediction time), energy drift, and downstream energy-based swing-up control. Across three seeds and four training-set sizes we find a nuanced picture rather than a single winner. The LNN, whose learned positive-definite mass matrix matches the true structure of these systems, gives the best accuracy and the lowest energy drift and enables the most successful swing-up. The GP is strikingly data-efficient and yields the most stable long-horizon rollouts despite modest one-step accuracy. The LGP combines these strengths, matching LNN accuracy while inheriting GP rollout stability. In contrast, the HNN—whose canonical-coordinate assumption is violated by the configuration-dependent inertia of these robots—is *worse* than the black-box MLP and diverges in the majority of rollouts. Only models with a usable energy function can drive energy-based control, and control quality tracks energy-model quality. Our results map out concrete conditions under which physics priors help: scarce data, long horizons and energy-sensitive tasks, and, crucially, a prior whose structure matches the system. Code and data are released for reproducibility.

Index Terms—physics-informed machine learning, learned dynamics, underactuated robots, Lagrangian neural networks, Gaussian processes, swing-up control.

I. INTRODUCTION

Learning the dynamics of a mechanical system from data is a building block of model-based control and reinforcement learning. Plain neural networks can fit one-step transitions very accurately, yet when their predictions are rolled out over long horizons they tend to drift and to violate basic physical invariants such as conservation of energy. Physics-informed machine learning (PiML) addresses this by baking structure—Lagrangian or Hamiltonian mechanics, positive-definite inertia, energy conservation—directly into the model

[1], [2]. The promise is better long-horizon behaviour and better sample efficiency.

In practice, however, it is rarely obvious *when* a physics prior is worth its added complexity. Priors can help, but a prior whose assumptions are violated by the real system can also hurt. Most papers introduce a single physics-informed model and show it beats a baseline on a favourable task; what is missing is a controlled, like-for-like comparison across several priors, several evaluation criteria, and several data budgets.

This paper provides such a comparison. We benchmark five dynamics models on two canonical underactuated systems and evaluate them along four complementary axes. Our contribution is not a new model but a *controlled empirical map* of when physics priors help, summarised by three findings:

- 1) **Matching structure matters most.** The LNN, which learns a positive-definite, configuration-dependent mass matrix—exactly the structure of these robots—dominates on accuracy and energy conservation and enables the best swing-up control.
- 2) **The wrong prior is worse than no prior.** The HNN, which assumes canonical coordinates with an identity mass matrix, is consistently worse than the black-box MLP and diverges in 60–87% of long rollouts.
- 3) **Uncertainty buys stability and data efficiency.** The GP is the most data-efficient model and produces the most stable long-horizon rollouts despite modest one-step accuracy; combining it with Lagrangian structure (the LGP) yields the best overall blend on the dynamics axes.

All experiments are reproducible from a released, configuration-driven pipeline.

II. RELATED WORK

Physics-informed neural dynamics. Deep Lagrangian Networks (DeLaN) [3] learn the mass matrix and potential of a Lagrangian and recover the equations of motion through automatic differentiation; Lagrangian Neural Networks [4] generalise this idea. Hamiltonian Neural Networks [5] instead

learn a Hamiltonian and integrate the symplectic flow, conserving energy by construction but requiring canonical coordinates. Symplectic and variational-integrator variants extend these ideas to control and to noisy data [6], [7]. Physics-informed neural networks more broadly embed governing equations into the loss [2], [1].

Gaussian-process dynamics. GPs provide calibrated uncertainty and strong data efficiency for model learning [8], [9], and underpin sample-efficient model-based control such as PILCO [10]. Hybrid schemes place GP priors on physically meaningful quantities or learn a GP residual on top of a parametric model.

Underactuated swing-up. The Acrobot and Pendubot are standard underactuated benchmarks; energy-shaping controllers swing them toward the upright equilibrium by regulating total energy [11], [12]. We use swing-up as a downstream task that stresses the *usefulness* of a learned model rather than only its predictive error.

Relative to this literature, we hold the data, integrator, training objective, and evaluation fixed across all five models, enabling a fair comparison rather than a single favourable demonstration.

III. SYSTEMS AND PROBLEM SETUP

We study two planar double pendulums with a single actuator. Both share the manipulator equation

$$M(q)\ddot{q} + C(q, \dot{q})\dot{q} + g(q) = Bu, \quad (1)$$

where $q = (q_1, q_2)$ are the joint angles, $M(q) \succ 0$ is the mass matrix, C collects Coriolis/centrifugal terms, $g(q) = \partial V / \partial q$ is the gravity vector for potential V , and B is the actuation matrix. The **Acrobot** is actuated only at the elbow ($B = [0, 1]^\top$) and the **Pendubot** only at the shoulder ($B = [1, 0]^\top$); the other joint is passive, which makes both systems underactuated. The state is $x = (q_1, q_2, \dot{q}_1, \dot{q}_2)$ and the control u is a scalar torque. We integrate the true dynamics with RK4 at $\Delta t = 10^{-2}$ s.

Given transitions (x_t, u_t, x_{t+1}) collected under random torques, the task is to learn a model $\hat{f}$ that predicts accelerations (hence the next state via the same RK4 integrator). We deliberately use a single integrator and a single one-step training objective for every model, so that differences in behaviour are attributable to the prior, not the training setup.

IV. THE MODEL ZOO

All models map (x, u) to a predicted state derivative and are rolled out with RK4. They differ only in their inductive bias.

Vanilla MLP (black box). A multilayer perceptron regresses the joint accelerations directly. No physical structure is imposed.

Gaussian Process (GP). An exact GP with an automatic-relevance-determination RBF kernel regresses each acceleration component from features $[\sin q_1, \cos q_1, \sin q_2, \cos q_2, \dot{q}_1, \dot{q}_2, u]$, with hyperparameters fit by marginal likelihood. The $(\sin, \cos)$ encoding respects

angle periodicity. To keep the exact GP tractable we cap the training set at 1000 points.

Lagrangian Neural Network (LNN). Following DeLaN [3], two small networks output a Cholesky factor of the mass matrix $M(q) = L(q)L(q)^\top + \epsilon I \succ 0$ and a scalar potential $V(q)$. The Coriolis term and gravity are obtained by automatic differentiation, and the acceleration is recovered from $M\ddot{q} = Bu - C\dot{q} - g$. The known actuation matrix B is supplied as a prior.

Hamiltonian Neural Network (HNN). Following [5], a scalar Hamiltonian $H(q, p)$ is learned and the symplectic flow $\dot{q} = \partial H / \partial p$, $\dot{p} = -\partial H / \partial q + Bu$ is followed, conserving H at zero input. As is common when only velocities are available, we use the identity-mass convention $p := \dot{q}$; the HNN therefore does *not* learn a configuration-dependent mass matrix.

Lagrangian-informed GP (LGP). Our hybrid uses an LNN as a physics mean and a GP to model the residual acceleration it leaves behind: $\ddot{q} = \ddot{q}_{\text{LNN}}(x, u) + \text{GP}_{\text{res}}(x, u)$. It keeps the strong Lagrangian structure while letting the GP correct what the structural model misses.

Both physics-structured networks (LNN, HNN) conserve their own learned energy to $\sim 10^{-7}$ over 200 steps even before training, confirming the correctness of the autograd-based equations of motion.

V. EXPERIMENTAL SETUP

Data. For each system we generate 8000 transitions from random short rollouts under uniform random torques. A fixed 20% tail is held out for one-step evaluation; the remaining pool is randomly subsampled to training budgets of 250, 1000, 4000, and 6400 transitions to probe data efficiency.

Training. Every model is trained to minimise a one-step, angle-aware mean-squared error between the RK4-integrated prediction and the recorded next state (GP-style models are instead fit by marginal likelihood). We use 120 epochs, Adam, gradient clipping, and three random seeds per configuration.

Evaluation axes. (1) *One-step MSE* on the held-out set. (2) *Valid prediction time*: the number of steps an autoregressive free-swing rollout tracks the truth before the per-step error exceeds a fixed threshold—a metric suited to chaotic systems, where raw long-horizon MSE saturates and stops discriminating. (3) *Energy drift*: the range of the true total energy along the model’s free rollout, which should be zero for a conservative system; we also report the fraction of rollouts that diverge. (4) *Swing-up control* (Section VI-B). All rollout evaluations use identical, fixed initial conditions across models for fairness.

VI. RESULTS

A. Prediction, Stability, and Energy (Axes 1–3)

Table I reports the three dynamics axes at the largest data budget, averaged over seeds; Fig. 1 shows how they vary with the amount of training data.

Accuracy and energy. The LNN is the most accurate model on both systems (one-step MSE below 10^{-3}) and has the

TABLE I

DYNAMICS AXES AT THE LARGEST TRAINING BUDGET ($n=6400$), MEAN OVER THREE SEEDS. ARROWS INDICATE WHETHER LOWER ($\downarrow$) OR HIGHER ($\uparrow$) IS BETTER. “DIV.” IS THE FRACTION OF FREE-SWING ROLLOUTS THAT DIVERGED.

System	Model	One-step MSE $\downarrow$	Valid steps $\uparrow$	Energy drift $\downarrow$	Div.
Acrobot	Vanilla MLP	3.0×10^{-3}	31.5	1.4×10^1	0%
	Vanilla GP	9.9×10^{-3}	72.0	7.5	0%
	Lagrangian NN	7.5×10^{-4}	68.8	4.8	0%
	Hamiltonian NN	1.3×10^{-2}	12.9	9.3×10^2	60%
	Lagrangian GP	9.1×10^{-4}	62.3	7.0	0%
Pendubot	Vanilla MLP	4.8×10^{-3}	35.6	1.3×10^1	0%
	Vanilla GP	6.9×10^{-3}	78.2	7.7	0%
	Lagrangian NN	8.9×10^{-4}	46.8	5.1	0%
	Hamiltonian NN	1.3×10^{-2}	17.6	9.9×10^2	87%
	Lagrangian GP	1.4×10^{-3}	64.1	5.4	0%

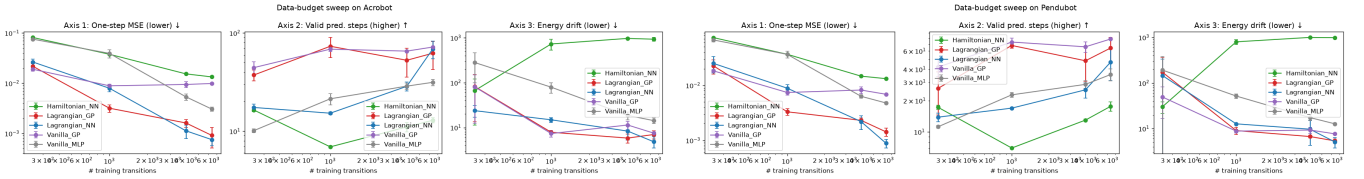


Fig. 1. Data-budget sweep on the Acrobot (left) and Pendubot (right). Each panel shows one dynamics axis versus the number of training transitions (log-log), averaged over seeds with standard-deviation error bars.

lowest energy drift, reflecting its matched physical structure. The black-box MLP is a competent middle performer.

Stability. On the chaotic free-swing task, the GP yields the *longest* valid prediction time on both systems—more than twice the MLP’s—despite its larger one-step error. Smoothing toward the GP prior appears to damp the error growth that destabilises the MLP. The LGP inherits much of this stability while retaining LNN-level one-step accuracy, giving the best overall balance: on the Pendubot it even exceeds the plain LNN in valid steps.

The wrong prior hurts. The HNN is the worst model on every dynamics axis. Its identity-mass assumption is badly violated by the configuration-dependent inertia of these robots, so it fits poorly even one step ahead and diverges in 60–87% of rollouts (its huge energy-drift values are an artifact of these divergences). This is a concrete case where an ill-matched physics prior is worse than no prior at all.

Data efficiency (Fig. 1). At the smallest budget (250 points) the GP and the Lagrangian models are already far more accurate than the MLP, whose advantage only appears with more data. Physics priors and GP smoothing both help most exactly when data are scarce.

B. Downstream Swing-Up Control (Axis 4)

We close the loop with an energy-shaping controller that pumps energy into the *true* plant using a model’s energy estimate,

$$u = k_e (E_{\text{top}} - E_{\text{model}}(x)) \dot{q}_{\text{act}} - k_p q_{\text{act}} - k_d \dot{q}_{\text{act}}, \quad (2)$$

clipped to a torque limit, where E_{top} is the upright energy under the same model (so any constant offset cancels) and q_{act} is the actuated joint. Only models that expose an energy

TABLE II

ENERGY-BASED SWING-UP. MEAN NORMALISED TIP HEIGHT AND SUCCESS RATE (SEED-AVERAGED). THE ORACLE USES THE TRUE ENERGY. THE MLP AND GP HAVE NO ENERGY FUNCTION AND CANNOT DRIVE AN ENERGY-BASED CONTROLLER (N/A).

System	Model	Mean tip height	Success
Acrobot	Oracle	0.97	100%
	Lagrangian NN	0.69	33%
	Lagrangian GP	0.62	25%
	Hamiltonian NN	0.50	0%
Pendubot	Oracle	0.96	100%
	Lagrangian NN	0.84	50%
	Hamiltonian NN	0.67	8%
	Lagrangian GP	0.61	17%

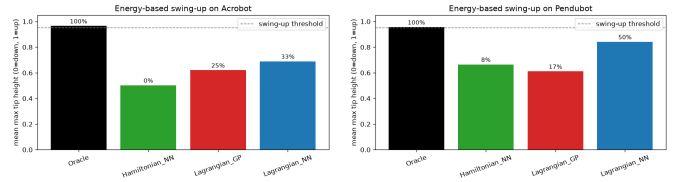


Fig. 2. Swing-up performance. Bars show mean tip height; the percentage above each bar is the success rate. Black-box models are absent because they provide no energy function.

function—the LNN, HNN, and LGP—can drive this controller; the black-box MLP and GP cannot, which is itself a result. We report the normalised tip height reached (0 down, 1 upright) and the success rate over fixed initial conditions, with a true-energy *Oracle* as the ceiling (Table II, Fig. 2).

The ranking on control follows energy-model quality: the LNN, with the most accurate learned energy, swings up best

among learned models (up to 50% success on the Pendubot), the LGP follows, and the HNN—whose energy is unreliable—largely fails. No learned model matches the Oracle, underlining that partial energy errors still cost control performance. This directly connects predictive quality to downstream usefulness.

VII. DISCUSSION: A MAP OF WHEN PHYSICS PRIORS HELP

Taken together, the four axes yield a consistent picture. *First*, physics priors help most when their structure matches the system: the LNN’s learned positive-definite, configuration-dependent mass matrix is the single most valuable ingredient here, driving its wins on accuracy, energy, and control. *Second*, an ill-matched prior can be actively harmful—the HNN’s identity-mass assumption makes it worse than a black box and unstable. *Third*, the benefit of any prior is largest in the low-data regime and on long-horizon, energy-sensitive tasks; at high data with only one-step accuracy in view, the black-box MLP is competitive. *Fourth*, uncertainty-aware models (GPs) buy rollout stability and data efficiency that are complementary to hard structural priors, and combining the two (the LGP) gives the best balance on the dynamics axes. No single model wins everything, which is precisely the practical guidance a practitioner needs.

VIII. LIMITATIONS AND FUTURE WORK

Our study is in simulation on two planar systems with a single, simple energy-shaping controller; absolute swing-up success is modest because the learned energy is imperfect and the controller is not re-tuned per model. The exact GP is capped in training-set size, and the GP shows little seed variance at the largest budget because its fit is deterministic there. Natural extensions include observation and process noise, per-model controller tuning or an LQR catch near the top, model-predictive control so that black-box models can also be evaluated downstream, additional systems (e.g. cart-pole), and sparse or deep-kernel GPs to scale the GP and LGP to larger budgets.

IX. CONCLUSION

We presented a controlled, reproducible benchmark of five dynamics models on two underactuated systems across four evaluation axes. Rather than crowning a single method, our results map out *when* physics priors help: most when the prior matches the system’s structure, when data are scarce, and on long-horizon, energy-sensitive tasks—while a mismatched prior can be worse than none. We hope the released pipeline serves as a reusable testbed for physics-informed dynamics learning.

REPRODUCIBILITY

Code, configurations, datasets, and the scripts that regenerate every table and figure are available at <https://github.com/amirhossein-fattahi/piml-underactuated>.

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